

DATA/CS 172: Python for Data Analysis

Intro to matplotlib

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Reminders and Announcements

- Program 2 out
 - Should be able to do now – uses your CSV, API, etc. skills
 - Have quite a while to work on it
- Keep on top of labs

Today

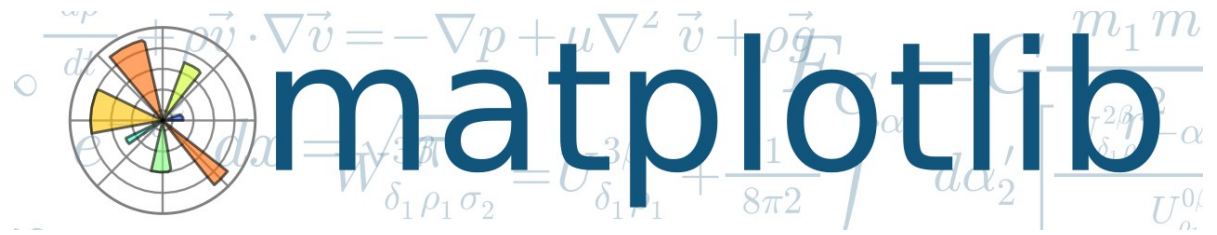
- Basics of data visualization

We've worked with many different kinds of data...

- Text Data
- More recently:
 - CSV Data
 - JSON Data
- But we have just been able to spit out basic facts about the data
- A big part of data science is how to communicate complex data to non-data-scientists....
 - We need **visualization**
 - Graphs, charts, etc.



matplotlib



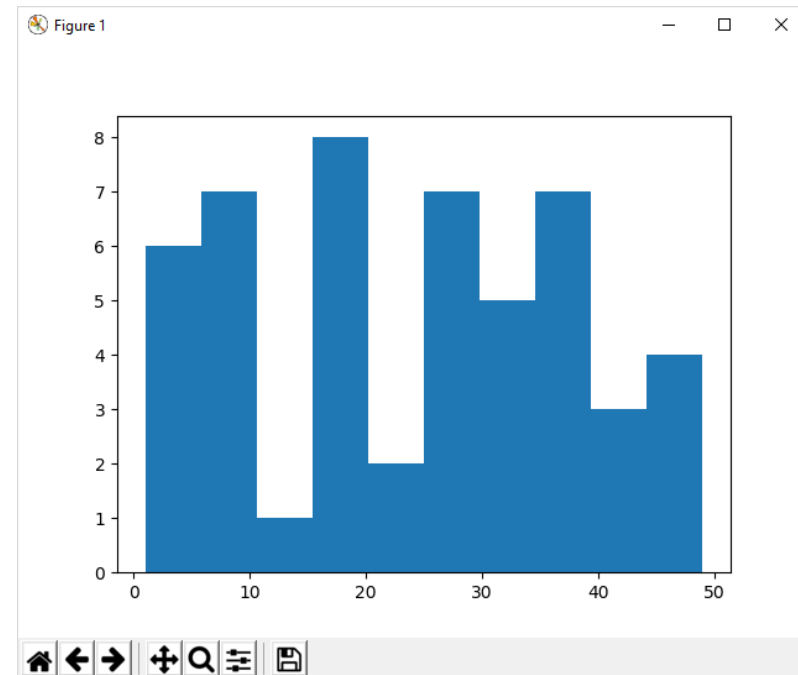
- Python plotting library
- Around since 2002, really started taking off when people started using Python for data science
 - There are other Python Plotting libraries
 - Matplotlib definitely still #1 in Python graphing
 - Can do many kinds of powerful plots
- Chapter 15 in Crash course book
 - Online documentation may be more helpful

Installing matplotlib

- Need to do this locally
- `pip install matplotlib`
- From local command line (cmd or Terminal)
- Test by running “`import matplotlib`” in interactive mode

A basic histogram

- Import matplotlib.pyplot as plt
 - So you don't have to type the name every time – very common
- `plt.hist(<list of values>)`



Showing the figure

- `plt.show()`
 - Pops up a window with the figure
 - Script pauses until window is closed
 - Later commands DO NOT execute

Changing number of bins

- `plt.hist(<numbers>, bins=<number of bins>)`

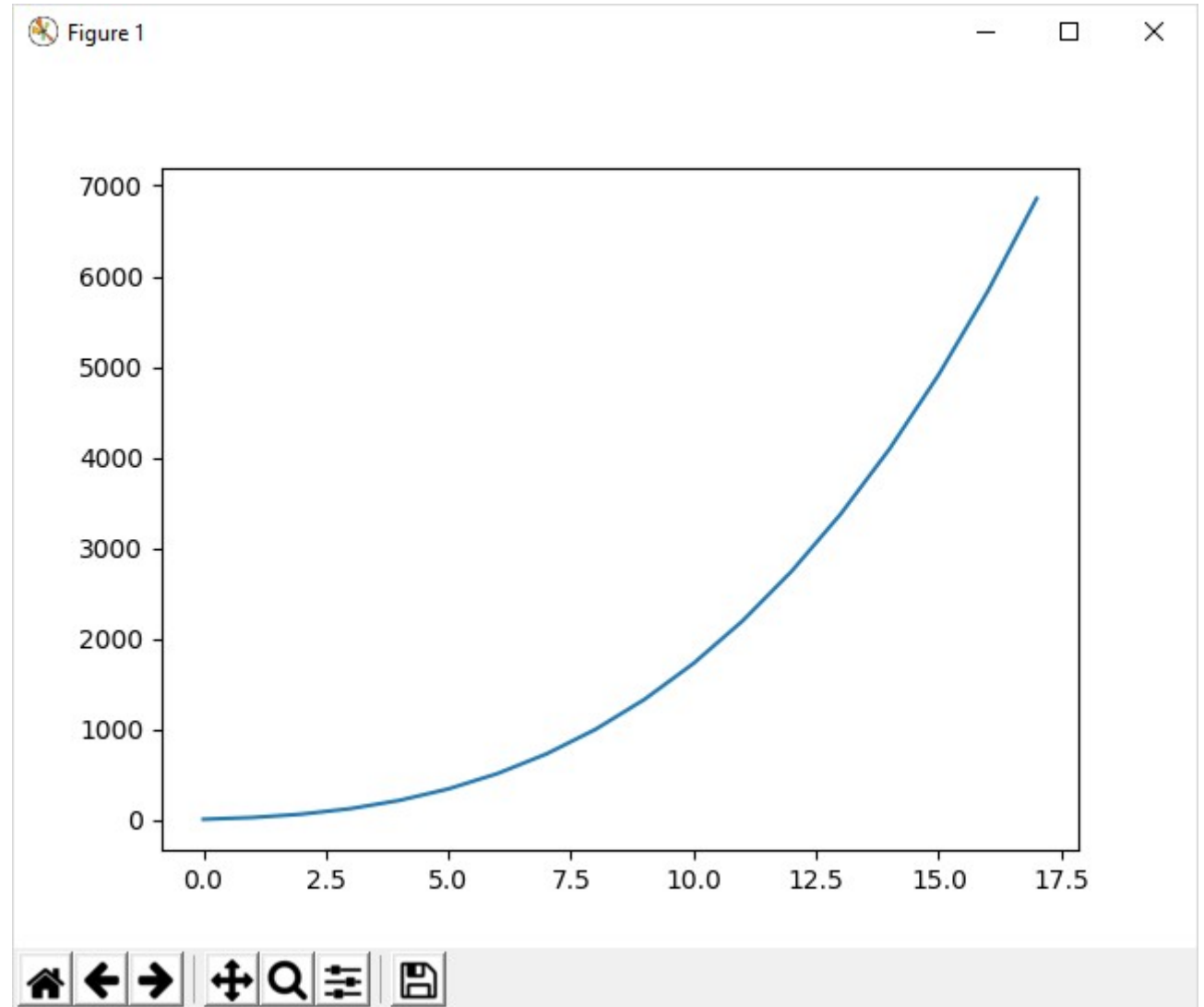


Adding axis labels

- `plt.xlabel("X Label")`
- `plt.ylabel("Y Label")`
- `plt.title("Title")`

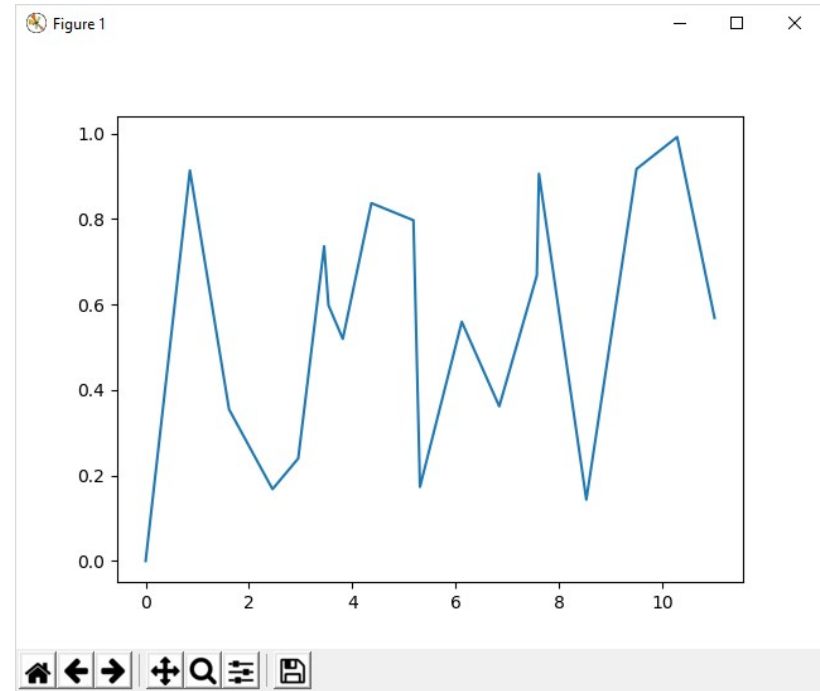
Next up: Line graph

- `plt.plot(yValues)`



X & Y line graph

- By default, xs appear at 0,1,2,3,4,5,6,7,8,9
 - Often not what we want!
- `plt.plot(xs, ys)`



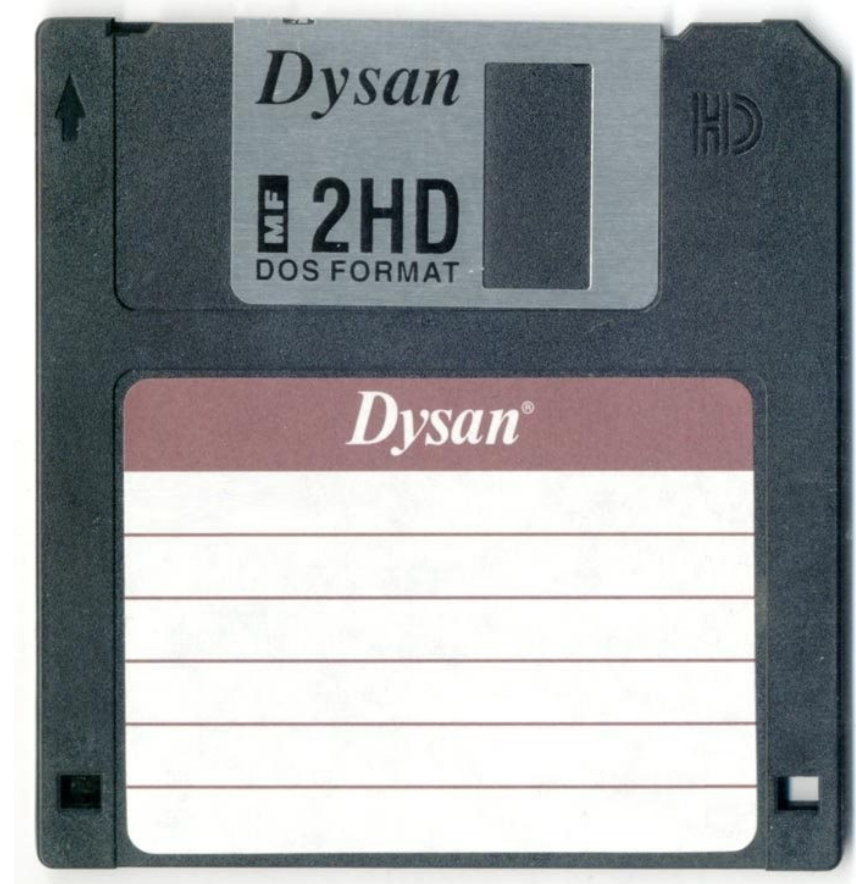
Changing axis ranges

- `plt.xlim([0, 100])`
 - Sets x from 0 to 100
- `plt.ylim([0, 10])`
 - Sets y from 0 to 10



Saving figures

- Can manually save by pressing the disk icon
 - Inconvenient
- `plt.savefig("filename.extension")`
 - Relies on file extension
 - Can save in rasterized formats (png, jpg)
 - Or in vector graphics (svg, pdf)
 - Infinitely zoomable/no pixelation
 - Often best for papers, reports, etc.



Lab out

- Make your first graph
 - Based on CSV data – refer to last Wednesday's lecture if you don't recall how to do this
- Friday: More matplotlib!